Antiviral Resistance

The specter of drug resistance hangs over the family of antiherpes drugs, as it does over all other classes of antimicrobials. However, clinically relevant resistance is almost exclusively observed in immunocompromised patients. The primary mechanism of acyclovir resistance involves the selection of viral mutants with defective or deficient thymidine kinase (TK) expression. Most TK-deficient mutants are somewhat attenuated in virulence in vivo.

Treating resistant HSV infection is challenging and requires two key steps. First, resistance must be accurately diagnosed. Many individuals who believe they are "resistant" to antiviral drugs do not actually harbor resistant virus. A common misconception is that antiviral therapy should completely prevent all recurrences indefinitely. Resistance should only be suspected in patients who continue to experience culture-proven outbreaks with unchanged frequency and severity, particularly when lesions fail to heal spontaneously.

When resistance is suspected, the virus should be isolated and tested for acyclovir sensitivity. These tests are costly but are now widely available through commercial reference laboratories.

Second, treatment options for confirmed resistant HSV are limited and suboptimal due to the lack of safe, convenient alternatives. **Foscarnet**, an organic analogue of inorganic pyrophosphate, inhibits replication of all known herpesviruses in vitro. Because it does not require activation by TK or other viral kinases, it remains effective against acyclovir-resistant HSV. However, foscarnet requires intravenous administration and is associated with significant adverse effects, including nephrotoxicity, electrolyte disturbances, anemia, and seizures. Furthermore, foscarnet-resistant herpesvirus strains have been reported.

Cidofovir, an acyclic nucleoside phosphonate, has broad-spectrum activity against DNA viruses, including herpesviruses. As a nucleotide analog, it does not depend on viral TK for intracellular activation, making it effective against TK-deficient mutants. Topical cidofovir has been successfully used to treat progressive herpetic lesions in cases of acyclovir resistance. Intravenous cidofovir is associated with substantial nephrotoxicity and requires concomitant saline hydration and probenecid to mitigate renal damage. There is a single report of acyclovir-resistant genital herpes responding to **imiquimod 5% cream**, though imiquimod has caused severe local inflammation in some patients with recurrent herpes labialis.

Prevention

Efforts to prevent HSV infection have been insufficient in curbing the rising prevalence of genital herpes. Complete abstinence from sexual contact prevents transmission, as evidenced by very low seroprevalence rates in populations such as cloistered nuns. Consistent condom use reduces transmission rates, although protection is incomplete.

Beyond public health measures, prevention strategies primarily focus on antiviral therapy and vaccine development, particularly for genital herpes.

Antiviral Therapy

Acyclovir, **famciclovir**, and **valacyclovir** significantly reduce both symptomatic and subclinical shedding of HSV-2. Viral shedding decreases from approximately 8% of days in placebo recipients to 0.3%–0.6% of days in those receiving antiviral therapy, as measured by viral culture.

In a randomized, placebo-controlled trial involving immunocompetent, heterosexual couples in stable relationships, **valacyclovir 500 mg once daily** reduced HSV-2 transmission by 48% and clinical disease in the susceptible partner by 75%. This regimen can be recommended for individuals concerned about transmitting HSV-2 to their partners, especially when combined with condom use.

However, the efficacy of antiviral therapy in reducing transmission among other populations—such as homosexual couples, non-monogamous individuals, immunocompromised persons, and those with asymptomatic HSV-2 infection—remains uncertain. Optimal regimens for these groups have not been clearly established.